

HUI “AMALIE” SHI

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SUMMARY

Experienced and results-oriented software engineer with strong quantitative, analytical and communication skills and a successful track record developing, integrating, and validating software solutions for the pharmaceutical and medical device industries.

INDUSTRIAL EXPERIENCE

Software Test Engineer Adaptix Ltd., Engineering

October 2022 - Present

- Architected and deployed a suite of 30+ enterprise-grade automation frameworks using **Django, PyTest, and PyWinAuto** to verify 3D medical imaging UIs and communication portals; modernized verification workflows and significantly increased test coverage for high-stakes, mission-critical healthcare systems.
- **Led end-to-end validation and verification** of design requirements for **veterinary and orthopedic X-ray systems**, ensuring compliance with **IEC 602304** and other international regulatory standards.
- **Authored and presented detailed technical test reports** to cross-functional teams, enabling resolution of **45+ software and hardware defects** through analytical troubleshooting, strong communication, and collaborative problem-solving.
- **Streamlined testing workflows and quality processes**, resulting in improved defect detection efficiency and faster release cycles.

Software Applications Engineer Informetric Systems Inc., Engineering

Feb 2021 - August 2022

- **Gathered, analyzed, and mapped user and process requirements into configurable automation logic for InfoBatch manufacturing systems, driving digital transformation and process optimization for pharmaceutical clients.**
- **Developed, tested, and validated 5 software updates and hotfixes** for InfoBatch and **managed the software release and product launch of InfoLog**, including setup of **VM-based test environments**, creation of **installation packages using Advanced Installer**, and design of **new automated test procedures**.
- **Provided high-level technical support** to resolve complex issues (e.g., HTML report formatting, delayed data uploads, printer integration) — ensuring **system uptime, customer satisfaction, and operational continuity**.
- Collaborated with **software developers, QA, and client engineering teams** to align requirements and deliver robust, production-ready solutions.

Pharmaceutical Engineer Aphenia Pharma Solutions, Engineering

Jan 2020 - Feb 2021

- **Authored and executed 30+ validation protocols and reports (IQ/OQ/PQ)**, commissioning **8+ automated systems** — including **blending tanks, labeling machines, water purification units, and serialization equipment** — under **FDA cGMP and engineering validation standards**.
- **Designed and optimized 5+ manufacturing and lab-scale processes** using **engineering design principles and data-driven analysis**, improving **process efficiency** and product consistency.
- **Implemented CAPA and preventive maintenance programs**, driving **equipment reliability, reduced downtime, and cost efficiency**.
- Collaborated with **R&D, QA, and manufacturing teams** to implement **engineering process improvements**, ensuring compliance with **regulatory and technical standards**.

PROJECTS

Workflow-based Automation of UI software Testing

2024

Enhanced Performance and Safety Verification for Medical Software:

- Developed and implemented automated testing for the software UI of Adaptix's compact, low-dose, 3D X-ray imaging systems by utilized **PyWinAuto** to automate user behaviors, performing image acquisitions thousands of times without manual intervention.
- Reduced manual labor and human errors by implementing continuous testing and verification processes and increased routine testing frequency, allowing for early detection of software errors and improving the robustness of the software development lifecycle.

InfoBatch Upgrade Project

2021

Successfully executed and validated the Software Upgrade of InfoBatch with Microsoft SQL:

- **Biogen:** Upgraded from version 3.0 to version 4.0 on the AWS cloud-based systems for two sites.
- **RE Mason:** Upgraded from version 2.0 to version 4.0 for two sites.

SKILLS

Mathematical Sciences, Data, and Analysis: Python, Django, HTML, CSS, Tailwind, Git, Javascript, Data Management, Data Mining, MATLAB, Reporting and Visualization, Microsoft SQL server, Statistical Analysis, Data Analysis, Database Management

Communications: Project Management, Process Improvement, Research and Development, Excel, R, Product Requirement Management, Jira

EDUCATION/ PROFESSIONAL CERTIFICATIONS

JOHNS HOPKINS UNIVERSITY, Whiting School of Engineering, Baltimore, Maryland

Master of Science in Biomedical Engineering:

Thesis: [The Use of 3D-Printed Phantoms For Evaluating CT Image Quality](#)

Dec 2019

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta

Bachelor of Science in Biomedical Engineering with Highest Honors

Thesis: [The Use of Implanted Intramuscular FES system for Ameliorating Foot Drop During Locomotion](#)

Dec 2016

PROFESSIONAL CERTIFICATIONS

ISTQB Certified Test Automation Engineer (CTAL-TAE) , BCS, the Chartered Institute for IT

May 2025

Project Management Professional (PMP), Project Management Institute (PMI)

Feb 2025

Cloud Web Development Bootcamp, HyperionDev

Dec 2024

ISTQB Certified Foundation Level 4.0, BCS, the Chartered Institute for IT

Nov 2024

Engineer-In-Training (EIT), Chemical Engineering Discipline, Maryland State Board for Professional Engineers

Aug 2021

Certified Six Sigma Yellow Belt (CSSYB), American Society for Quality (ASQ)

Sept 2020

PUBLICATIONS AND CONFERENCES

[1]. **Shi, H.**, Woods, D., Egan, C., & Wilson, A. (2026, May). *Building Scalable and Automated Verification Tools for Healthcare Imaging Systems*. *Medical Informatics Europe (MIE 2026)*, Genova, Italy.

[1]. **Shi, H.** (2025, March). *Selenium Conference 2025: Talk - The Ultimate Swiss Army Knife of UI Software Testing: Accelerating Workflow-Based Automation*. Selenium Conference 2025, Valencia, Spain

[2]. **Shi, H.** (2024, July). *RSLondonSouthEast 2024: Abstracts – Research Software London*. RSLondonSouthEast, London, Imperial College London, Presenting a Workflow of Automating UI Software Testing and Reviewing a Commercial Case Study. <https://rslondon.ac.uk/rslondonse2024/abstracts/#p2>

[3]. **Shi, H.** (2024b). Workflow-based Automation of UI software Testing - A Review of a Commercial Case Study. *Zenodo*. <https://doi.org/10.5281/zenodo.12605845>

[4]. **Shi, H.**, Gang, G., Li, J., Liapi, E., Abbey, C., & Stayman, J. W. (2020). Performance assessment of texture reproduction in high-resolution CT. *SPIE Conference Proceeding*, 25. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7643885/>

[5]. **Shi, H.**, Lyle, M., Turtill, C., Nichols, R., *Positive force feedback may ameliorate muscle weakness*, Society for Neuroscience's 47th annual meeting

[6]. Lyle, M., **Shi, H.**, Anderson, H., Rapsas, B., *Behavioral adaptations during downslope walking after cross-reinnervation of medial gastrocnemius and the pretibial flexors*, Society for Neuroscience's 47th annual meeting

[7]. Li, J., Gang, G., **Shi, H.**, Stayman, JW., 3D-printed Texture Phantoms for Assessment of High-Resolution CT, American Association of Physicists in Medicine (AAPM) 2019 annual meeting